AAE 451 Aircraft Design – Spring 2026 Syllabus

Course information

Course number and title: AAE 45100 – 003 Aircraft Design
CRN: 10067 (lecture), 57509 (laboratory)
Meeting days and times:

- Lectures: TR 10:30 AM - 11:20 PM, WALC B074, instructional modality: face-to-face
- **Laboratories:** T 1:30 AM – 4:20 PM, PHYS 223, instructional modality: face-to-face, the lab sessions will be used for individual activities, tutorials for the computational tools, meetings with TAs, and design team meetings
- **Other:** Two evening sessions for design review presentations to off-campus review panelists (see the course calendar)

Course credit hours: 3.0
Prerequisites: Senior standing in AAE. Undergraduate level AAE 33400 Minimum Grade of D- and Undergraduate level AAE 34000 Minimum Grade of D- and Undergraduate level AAE 35200 Minimum Grade of D- and Undergraduate level AAE 36400 Minimum Grade of D- and Undergraduate level AAE 40000 Minimum Grade of D- [may be taken concurrently] and (Undergraduate level AAE 33900 Minimum Grade of D- or (Undergraduate level AAE 43800 Minimum Grade of D- [may be taken concurrently] and Undergraduate level AAE 43900 Minimum Grade of D- [may be taken concurrently]))

Course description

Official course description in the university catalog: Senior students perform a team-based aircraft design, requiring the application of the education and skills developed in the aerospace curriculum. Aircraft mission requirements include engine cycle selection and airframe/engine integration, performance, stability and control, structures, human factors, avionics, sensors, and manufacturing processes. The teams present oral and written reports on their designs.

Learning Resources, technology, and texts

Required textbook: Raymer, D., *Aircraft Design: A Conceptual Approach*, Sixth Edition, AIAA Education Series, 2018.

Dan Raymer is a Purdue graduate who wrote this book in a conversational style, making it easy to read and understand. The content draws from Dan's experiences working in aircraft design for major aircraft companies and his continuing experiences as a consultant and instructor of aircraft design courses. We expect you will want to keep this text after you are finished with AAE 451.

Obviously, the most recent edition has the most updated content, including an entire chapter on electric aircraft, but slightly earlier versions of the text will also be useful.

We have made this a required text, so that students have a consistent resource that covers material not explicitly discussed during class lecture. We expect students to read and apply approaches from this textbook for many aspects of the course, particularly in the development of a sizing code for their aircraft.

Suggested reading: Nicolai, L., and Carichner, G., *Fundamentals of Aircraft and Airship Design: Volume 1 - Aircraft Design*, Revised Edition, AIAA Education Series, 2010.

Lee Nicolai and Grant Carichner have a cumulative of 85 years working in aircraft design related areas. Both have spent most of their industry careers in the Lockheed Martin Skunk Works. The text contains relevant information for AAE 451. There are many similar themes to Raymer's text, but often approaches for analysis and design predictions differ from Raymer's text making this a nice complementary suggested text. This excellent book is also easy to read and understand.

Management system: Brightspace is our course management system. You can access the course website at **Brightspace**. It is strongly suggested that you explore and become familiar not only with the site navigation, but with content and resources available for this course. The material will be updated each week. Please check it regularly.

Lecture handouts: For each discussion or lecture topic, I will provide a set of handouts in PDF format that you will need to download from Brightspace, print (or download to your device, assuming you can "write" onto the handout), and have with you as you watch the class lecture. These handouts are intentionally incomplete; we will add information to these slides during the lecture. The idea is to have you participate by taking notes during the lecture because the activity of taking notes is associated with better retention of the material. However, with some of the material already provided, you should not rush to keep up with the notes and can listen more intently to the lecture. These notes are copyrighted by the course's current and past instructors and are for students registered in AAE 45100. You may not sell or otherwise distribute these notes.

Software / web resources: Students are expected to have access to and be familiar with the standard software using in engineering, including word processing, spreadsheets, presentation (e.g., PowerPoint) and computation (e.g., MATLAB). Students are free to choose the software and operating system that works best for their project team. Teams are encouraged to set up a common cloud storage area (e.g., OneDrive or Google Drive) to share work among members. Selection of the appropriate tools, including software, is an important part of the design process. Links to suggested software will be provided on Brightspace and introduction to them will be covered in the laboratories.

Learning outcomes

On completing this course, the student shall be able to:

1. Identify and formulate the design requirements for an aircraft
2. Conduct a set of trade studies to make decisions about the aircraft design
3. Describe an aircraft design that is likely to meet the set of requirements
4. Use modern engineering tools to complete the design process
5. Apply engineering knowledge and skills acquired in previous classes to the design problem
6. solve problems as part of a team
7. effectively communicate by giving oral design reviews and writing design reports/proposals

Assignments and evaluations

Design project assignment

This course will enable students to define an aircraft concept, beginning with a basic opportunity description. Students will work in teams to identify customers, formulate a basic business case, establish a basic concept of operations, develop the system requirements, generate concepts, conduct trade studies and sizing to determine the most promising aircraft design, and pursue conceptual design of the aircraft, with as much detail as time permits. The course will try to replicate industry or government conceptual aircraft design studies as closely as possible in a 15-week semester.

The topic of the Aircraft Design class for the spring 2026 semester will align with the 2025-2026 AIAA Undergraduate Team Aircraft Design RFP.

Graded items

Weekly assignments: The course is structured around a series of weekly assignments that serve as the technical building blocks for your team's aircraft design. To support your progress, the teaching team will provide detailed feedback on these submissions to help you refine your work as the semester progresses. If an assignment receives a grade below 90%, your team may revise and resubmit the work based on the provided feedback. Please note that the maximum grade achievable on any resubmitted assignment is capped at 90% to encourage high-quality initial submissions while still providing a pathway for everyone to master the core material.

Participation: You can earn this point by attending class and presentations and by responding to the daily attendance survey. To earn the maximum participation points, you should miss no more than two sessions.

ABET report: All students in AAE 45100 are required to complete an ABET Individual Reflection (aka: ABET report). This assignment involves answering several questions about your design experience during the course. You will type your responses directly into a prompt on Gradescope. While there are no right or wrong answers, your responses will be graded on their completeness. We anticipate that this activity will take between 30 minutes and an hour to complete.

Poster Presentation: During the final week of classes, teams will present their work to a panel of external judges. These presentations will last 10 minutes, and all team members are required to attend in business casual attire. The teaching team will handle the printing for you the day before the session. To ensure no text or images are cut off during the process, please ensure your poster file has at least a **0.22-inch (6mm) margin** on all sides.

RED Team Review: The RED Team Review is a formal peer review session where you will practice giving and receiving professional design critiques. During these sessions, you will be graded on two specific areas: the quality and usefulness of the feedback you provide as an audience member, and how professionally your team receives and responds to the feedback you are given. To ensure you get the high-level guidance needed for your project, I or a TA will also be in the room to provide "expert feedback" once the peer review portion is finished. This structure is designed to keep everyone engaged and ensure you leave the session with a clear, actionable path forward for your design.

Design deliverables:

There are four major "deliverables" for this class (as the semester proceeds, time available may dictate some changes in these deliverables):

1. A presentation with annotations for the System Definition Review (SDR)

2. A presentation with annotations for the Conceptual Design Review (CoDR)
3. A detailed and representative CAD model of the final version of the project (CAD)
4. A final written document following the requirements defined in the RFP document for the AIAA competition

The relative importance increases with each deliverable, reflecting that the content of each becomes more comprehensive. These design deliverables are analogs to mid-term exams in a more traditional class; attendance for the entire session in which your team presents is expected.

Report-based grade and peer evaluation

We will use the CATME methodology for peer evaluation, grading, and group formation. (https://info.catme.org/features/peer-evaluation/)

The instructor will assign a base grade to the team for each **group** deliverable. Because this is a team project, there is often considerable concern about the "unfairness" of assigning a single grade to all team members, especially when some group members feel they have done a disproportionately large share of the work. To address some of these issues, I will use a "Peer Evaluation Factor" approach to assign the grades for the course, relying upon a peer evaluation process. **The peer evaluation factor distribution should distinguish students who contributed positively to the team effort from those who did not.**

For each deliverable, each team member will complete a peer evaluation survey using CATME that describes their perspective of how his / her teammates contributed to the deliverable. Students will provide statements justifying their evaluations. The instructor will compile these to determine the peer evaluation factor. This approach to grading group assignments allows irresponsible team members to abuse them. Because of this, the **instructor retains ultimate authority over assigning the final grades to individual students.**

Design review presentations

The design review presentations will be presented to a panel of engineers via web meeting (Webex or similar). Making presentations to "real, practicing" engineers involved with aircraft design adds to the relevance of the capstone aircraft design class experience, provides practice in creating these kinds of professional-level presentations, and allows the panelists to provide feedback both about the approach to the design project and about the presentation style, format, and delivery. See the project page for additional information about the review panels.

The instructor and teaching assistant(s) will not formally review the presentations before the teams present them to the review panelists. This helps enhance the impact of the review panelists' feedback. The instructor and teaching assistant(s) will take notes during the presentations and provide written feedback to the teams.

Teaching team grading of the design review reports/presentations

The written report will be graded by the instructor, teaching assistant(s), and, if one is assigned, the grader. They will use a basic rubric to assess how well the reports address several different categories; these will align with the completion and quality of the activities required for each deliverable. Each report will receive a numerical score from the teaching team using this rubric. The team will also receive feedback from the teaching team while grading the report.

Course grades and grading scale

The final course grades will use the following weighting:

Graded item	Contribution to total course score
Weekly Assignments	25%
RED team reviews	10%
System definition presentation	10%
Conceptual design presentation	15%
Cad Model	10%
Final Report Document	15%
Poster Session	5%
ABET Report	5%
Participation	5%

After computing the scores for each student using the CATME peer evaluation factor for the design review/report scores, the following table converts the numerical percentage score.

The instructor intends to use this straight-scale approach to assigning final grades for the course, including the use of plus/minus modifiers (the instructor reserves the right to modify the grading scale, and any changes will be announced in class and on Brightspace):

Score	Grade	Score	Grade	Score	Grade	Score	Grade
98 to 100%	A+	88 to 89%	B+	78 to 79%	C+	68 to 69%	D+
93 to 97%	A	83 to 87%	B	73 to 77%	C	63 to 67%	D
90 to 92%	A-	80 to 82%	B-	70 to 72%	C-	60 to 62%	D-

Note on rounding up: I reserve the right to round or not round final grades as I see fit. In general? I round up. Just don't email me desperately a few days/hours before I have to turn in grades, begging me to round up. That will most likely cause me to round down.